

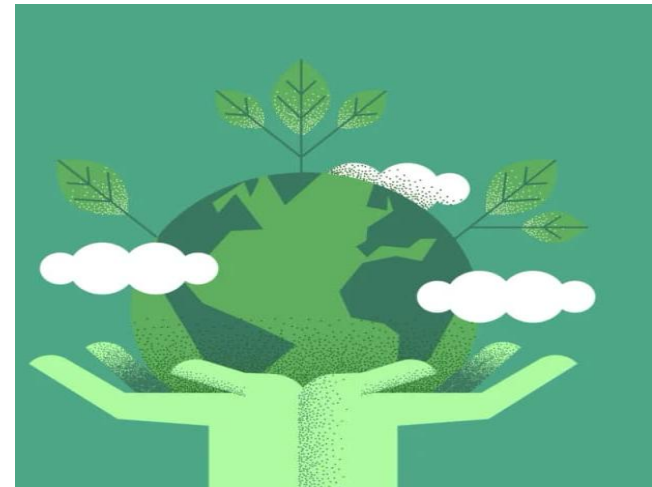


DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING

A Course on
INTRODUCTION TO SUSTAINABLE ENGINEERING



By
DEPARTMENT OF MECHANICAL ENGINEERING

OVERVIEW

- **Foundations of Sustainable Engineering** – Core principles of sustainability in engineering, systems thinking, lifecycle perspective, circular economy, product stewardship, and alignment with SDGs.
- **Energy & Resource Efficiency** – Identifying energy loss points, applying conservation strategies, optimizing designs, and improving material efficiency through sustainable sourcing and minimal embodied energy
- **Green Design & Manufacturing** – Designing for environment, reuse, disassembly, and waste minimization; adopting low-emission, efficient manufacturing processes with sustainable material choices
- **Sustainable Mobility & Infrastructure** – Light weighting, material substitution, structural optimization, and sustainable construction practices balancing safety, performance, and environmental impact
- **Digital Tools & Responsible Innovation** – Using LCA, AI, and predictive analytics for design optimization, material selection, and maintenance; ensuring ethical compliance with EPR, RoHS, and SDGs.

Course Learning Objectives:

This course will enable students to:

1. Introduce the foundational concepts of sustainable engineering and its role in addressing environmental and resource challenges.
2. Develop the ability to analyze energy and material flows in engineering systems and propose efficient conservation strategies.
3. Apply green design and sustainable manufacturing principles for minimizing environmental impacts across product life cycles.
4. Integrate sustainable approaches in mobility, infrastructure, and construction while balancing performance, safety, and environmental considerations.
5. Utilize digital tools, lifecycle assessment, and responsible innovation frameworks in making ethical and sustainable engineering decisions.

Course Outcome	Description
At the end of the course the student will be able to:	
1	Analyze engineering systems from a sustainability perspective and evaluate lifecycle and circular economy principles for improved system design.
2	Assess energy flows and material usage in complex engineering systems to propose efficient conservation and resource optimization solutions.
3	Design products and manufacturing processes that minimize emissions, waste, and environmental impact throughout the product lifecycle.
4	Evaluate sustainable mobility and infrastructure solutions by integrating material selection, structural optimization, and durability considerations.
5	Apply digital tools, lifecycle assessments, and AI-supported methods to optimize system performance while ensuring compliance with ethical standards and sustainability regulations.

TEXT BOOKS :

1. David T. Allen, David R. Shonnard, *Sustainable Engineering: Concepts, Design and Case Studies*, 2nd Edition, Pearson Education, 2012.
2. Bhavik R. Bakshi, *Sustainable Engineering: Principles and Practice*, Cambridge University Press, 2019.

REFERENCES :

1. David Dornfeld (Ed.), *Green Manufacturing: Fundamentals and Applications*, Springer, 2013.
2. Michael F. Ashby, *Materials and the Environment: Eco-Informed Material Choice*, 2nd Edition, Elsevier, 2012.



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UNIT-1

Foundations of Sustainable Engineering

SUSTAINABLE
ENGINEERING



Definition:

- The term sustainability has a **multidisciplinary use and meaning**.
- In dictionaries, sustainability is typically described by many sources as a capability of a system to endure and maintain itself.
- Various disciplines may apply this term differently.

In history of **humankind**, the concept of **sustainability was connected to human-dominated ecological systems from the earliest civilizations to the present**



- ❑ Sustainability has always been tied to how humans interact with and depend on nature—not just in modern times, but all throughout history

HISTORY OF SUSTAINABILITY



Civilization / Period	Dependence on Ecology	Sustainability Practice / Issue	Outcome / Challenge
Mesopotamia (Sumerians, Babylonians)	Agriculture thrived due to irrigation from the Tigris and Euphrates rivers	Issue: Over-irrigation without drainage caused soil salinity	Decline in agricultural productivity contributed to weakening of city-states
Indus Valley Civilization (Harappa, Mohenjo-Daro)	Advanced urban planning, water management, and agriculture	Practice: Built sophisticated drainage systems and seasonal crop cycles	Collapse linked to climate change (drought, river shifts) reducing water and output
Ancient Egypt	Sustained by annual Nile floods replenishing soil nutrients	Practice: Developed basin irrigation to use floodwaters efficiently	Achieved long-term ecological balance, lasting millennia
Maya Civilization (Mesoamerica)	Relied on forest resources, maize agriculture, and rainfall	Issue: Deforestation and soil erosion damaged land	Combined with droughts, stresses contributed to societal collapse
Ancient China	Depended on fertile river valleys (Yellow and Yangtze)	Practice: Terrace farming and crop rotation conserved soil and water	Population pressures caused famines and unrest when land use exceeded limits
Roman Empire	Agriculture and forests provided food, timber, and fuel	Issue: Deforestation, soil depletion, and overgrazing	Food supply collapse and ecological strain weakened empire
Industrial Revolution (18th–19th C.)	Shifted from local resources to global exploitation (coal, oil, metals, colonies)	Issue: Large-scale environmental degradation (pollution, deforestation, fossil fuels)	Sparked modern sustainability concerns: climate change, biodiversity loss, resource depletion

- ❑ Since the **1980s**, *sustainability* as a term has been used more in the sense of **human sustainability on planet Earth**
- ❑ This leads us to the concept of **sustainable development** which is defined by the Brundtland Commission of the **United Nations** (March 20, 1987) as follows:

Definition:

"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

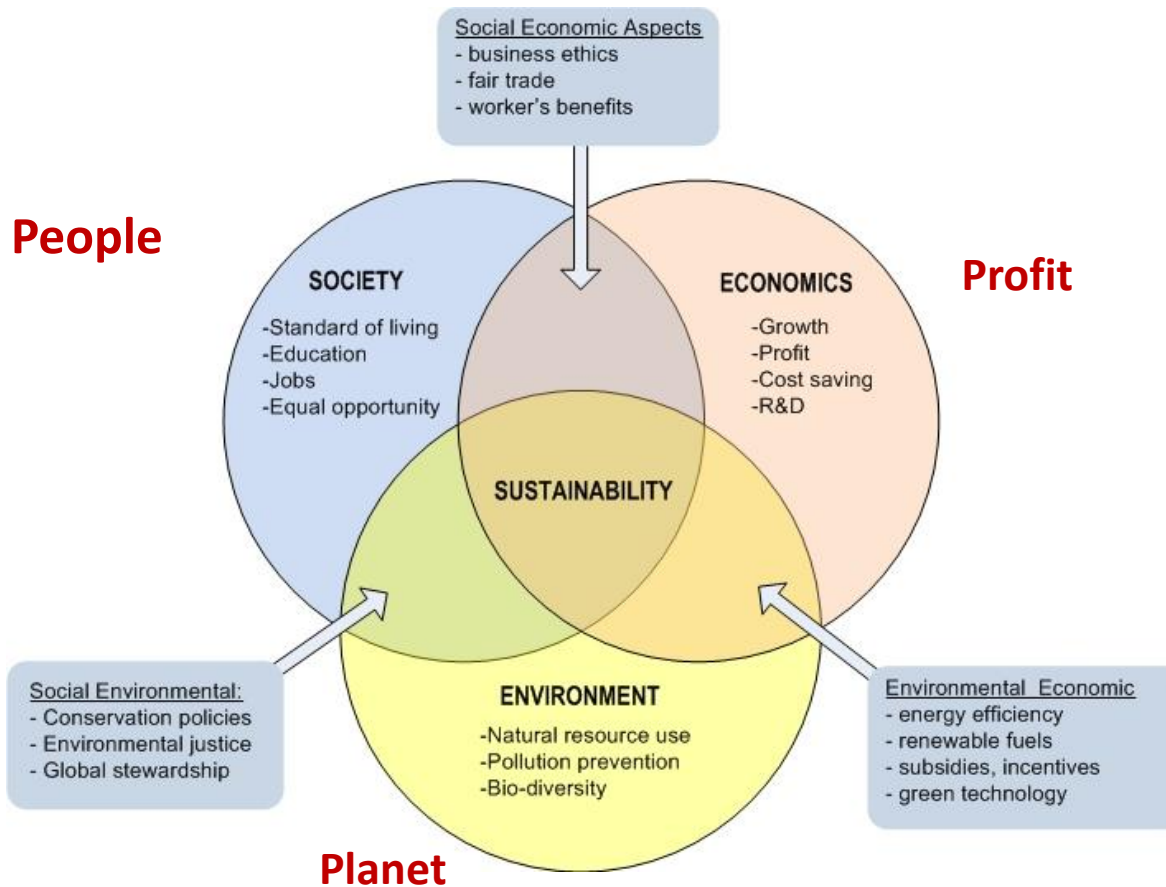
[World Commission on Environment and Development. 1987]

Sustainable engineering is the practice of designing, building, and maintaining systems and products in a way that minimizes environmental impact and resource depletion, while also considering social and economic factors. It aims to meet present needs without compromising the ability of future generations to meet their own needs.

<https://www.youtube.com/embed/rmQby7adocM>

Three Pillars of Sustainability

- ❑ Sustainable development involves **environmental, economic, and social aspects.**
- ❑ For a particular process to be sustainable:
 - ❑ It should not cause irreversible change to the environment,
 - ❑ It should be economically viable, and
 - ❑ It should benefit society.



Sustainability Engineering Core Principles:

- ❑ **Environmental Responsibility:** Minimizing pollution, waste, and resource consumption.
- ❑ **Social Equity:** Ensuring that engineering **solutions benefit all members of society**, including vulnerable communities.
- ❑ **Economic Viability:** Considering the **long-term costs and benefits** of engineering solutions.
- ❑ **Resource Management:** Promoting the **use of renewable resources** and efficient resource utilization.
- ❑ **Life Cycle Assessment:** Evaluating the environmental impact of **a product or system throughout its entire life cycle**, from raw material extraction to disposal.



❑ Sustainability Approaches in Engineering

Sustainability Approaches in Engineering

Traditional Engineering	Sustainable Engineering
Considers the object or process	Considers the whole system in which the object or process will be used
Focuses on technical issues	Considers both technical and non-technical issues synergistically
Solves the immediate problem	Strives to solve the problem for infinite future (forever?)
Considers the local context	Considers the global context
Assumes others will deal with political, ethical, and societal issues	Acknowledges the need to interact the experts in other disciplines related to the problem

- ❑ There **have been multiple attempts by academic and industrial institutions** to formulate sustainable engineering principles.
- ❑ All of them fall **within the triangle** with Environmental, Social, and Economic values as cornerstones.
- ❑ The overarching goal is to generate a **balanced solution** to any engineering problem.
- ❑ If an engineering project benefits one of these three aspects but ignores the others, we have a lopsided system which creates tension, instability, and new problems in the long run.

Challenges

Environmental Challenges

- ❑ **Climate Change:** Rising global temperatures, extreme weather events, sea level rise.
- ❑ **Pollution:** Air (industrial emissions), water (effluents, plastics), soil contamination.
- ❑ **Biodiversity Loss:** Habitat destruction, overexploitation, invasive species.
- ❑ **Waste Generation:** Growing volumes of hazardous and non-biodegradable waste.

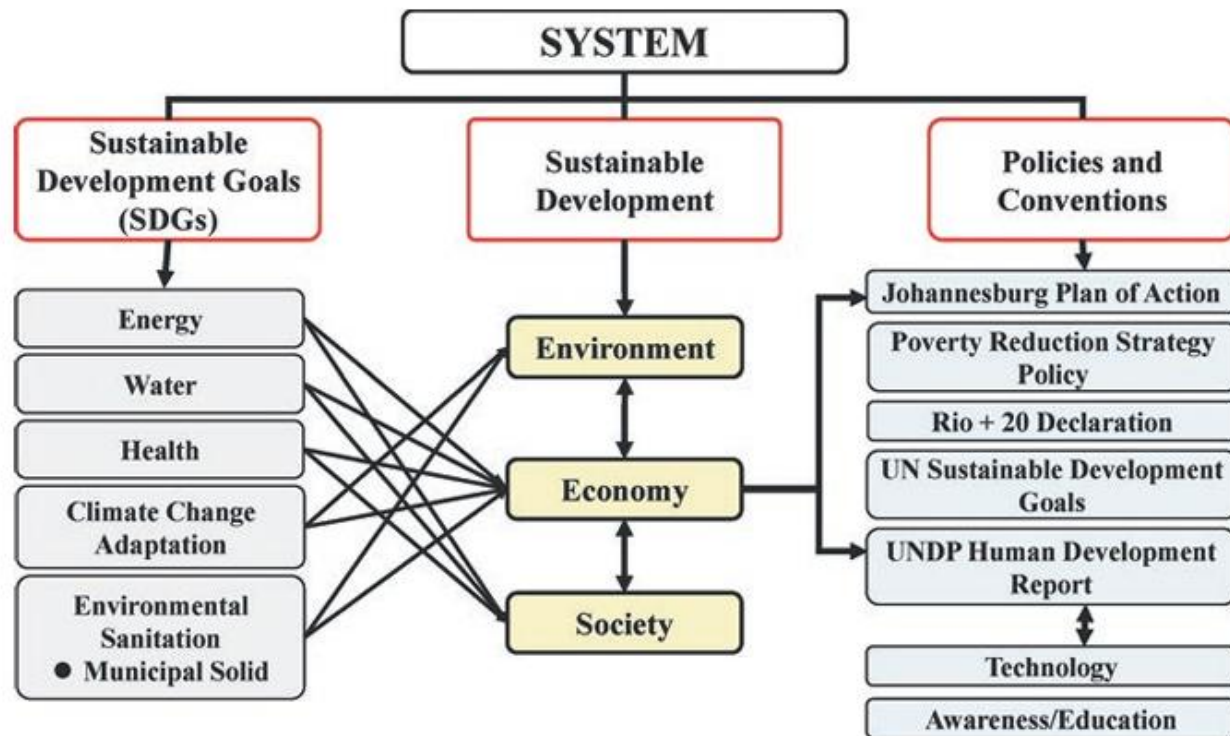
Resource Challenges

- ❑ **Resource Depletion:** Over-extraction of minerals, fossil fuels, forests.
- ❑ **Water Scarcity:** Increasing demand vs limited freshwater supply.
- ❑ **Energy Crisis:** Reliance on non-renewable energy sources.
- ❑ **Unsustainable Consumption:** Overuse of materials and goods beyond regeneration capacity.

Systems thinking in design and analysis

- To build the contextual framework for applying the sustainability principles, we need to develop some background in systems. We often hear terms like “**systems thinking**” or “**systems approach**”
- In some cases, to initiate a sustainable and long-lasting change, we need to change the “system” rather than trying to change the final result.
- A system is an interconnected set of elements that is organized in a way that achieves a purpose. Three distinct entities of any system are elements, interconnections, and purpose (or function).
- System’s properties and behaviours
 - resiliency,
 - self-organization,
 - self-repair, and **eventually - sustainability.**

- ❑ **Systems thinking** is crucial for sustainable design and analysis because it emphasizes understanding the interconnectedness and interdependencies within complex systems.
- ❑ By considering the broader context, including environmental, social, and economic factors, designers can create solutions that are more effective, resilient, and equitable.



Developmental targets and their interrelation to sustainable development

Lifecycle perspective

- ❑ **Life Cycle Assessment (LCA)** is a comprehensive method for evaluating the environmental impacts of a product or service throughout its entire life cycle, from raw material extraction to disposal.
- ❑ It's a "**cradle-to-grave**" analysis that aims to identify and quantify environmental burdens associated with each stage of the product's life.
- ❑ By assessing these impacts, **LCA helps businesses, policymakers, and consumers make more informed decisions about product design, resource use, and waste management, ultimately promoting sustainability.**



<https://www.youtube.com/watch?v=SMnioyRxWNk>

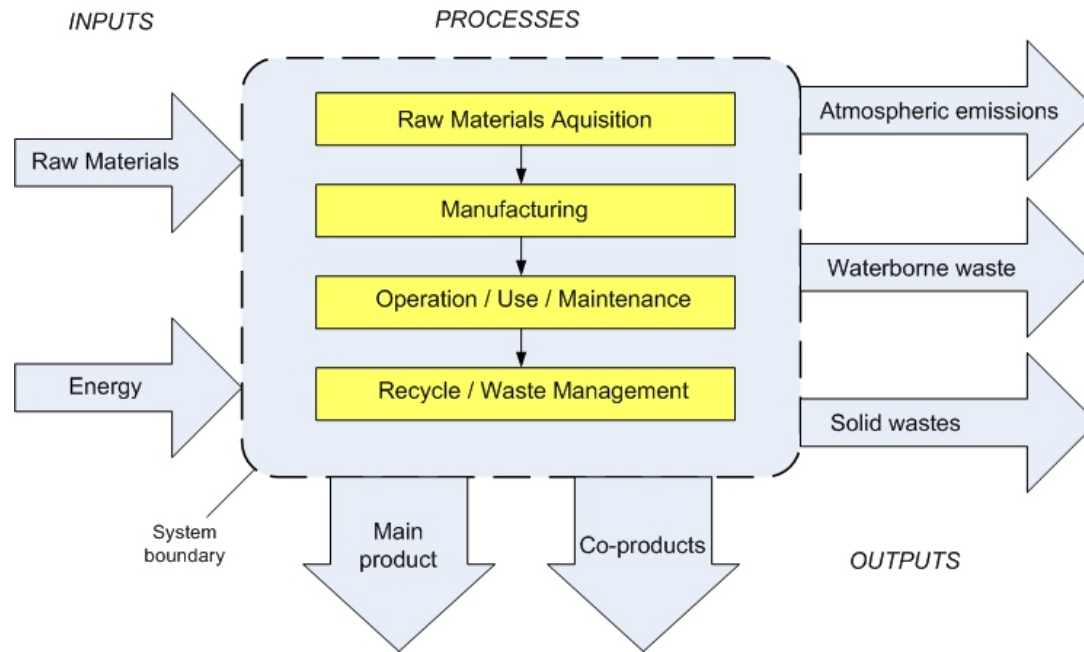
Life Cycle Assessment (LCA)

- ❑ LCA assesses the environmental impact of a product or service from **raw material extraction to disposal or recycling.**
- ❑ It considers all stages of the product's life, including **production, distribution, use, and end-of-life.**
- ❑ LCAs help identify areas where environmental impact can be reduced.
- ❑ LCA is a standardized process following ISO 14040 and 14044



International Organization for Standardization

The main stages and typical inflows and outflows considered in lifecycle assessment



- ❑ This diagram is based around a **box-shaped system** that includes **four processes** and is surrounded by a **System Boundary**. The four highlighted stages are:
 - ❑ **Raw Materials Acquisition**
 - ❑ **Manufacturing**
 - ❑ **Operation/Use/Maintenance**
 - ❑ **Recycle/Waste Management**

LCA Limitations

- ❑ LCA thoroughness and accuracy will depend on the availability of data;
 - ❑ gathering of data can be problematic; hence a clear understanding of the uncertainty and assumptions is important.
- ❑ Classic LCA will not determine which product, process, or technology is the most cost-effective or top-performing;
 - ❑ therefore, LCA needs to be combined with cost analysis, technical evaluation, and social metrics for comprehensive sustainability analysis.
- ❑ LCA models are suitable for relative comparisons, but may be not sufficient for absolute predictions of risks.

ISO 14000: **Environmental Management Standards**

- ❖ It provides organizations with **practical tools** to manage their **environmental responsibilities**.

- ❖ **Focus:**
 - ❖ Helping organizations reduce environmental impact,
 - ❖ comply with regulations, and
 - ❖ continuously improve environmental performance.

Objectives of ISO 14000

1. **Minimize environmental footprint** of an organization's processes, products, and services.
2. **Comply with laws and regulations** related to the environment.
3. **Prevent pollution and reduce waste.**
4. **Improve resource efficiency** (energy, water, raw materials).
5. **Promote sustainability** in business operations.

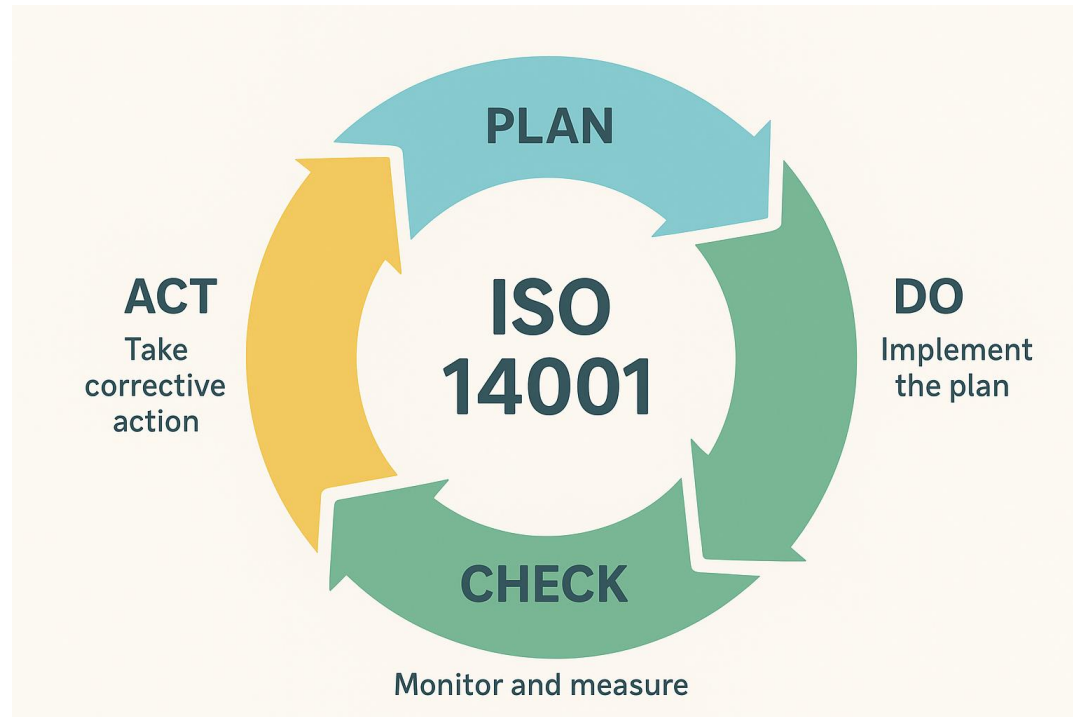
Objectives of ISO 14000



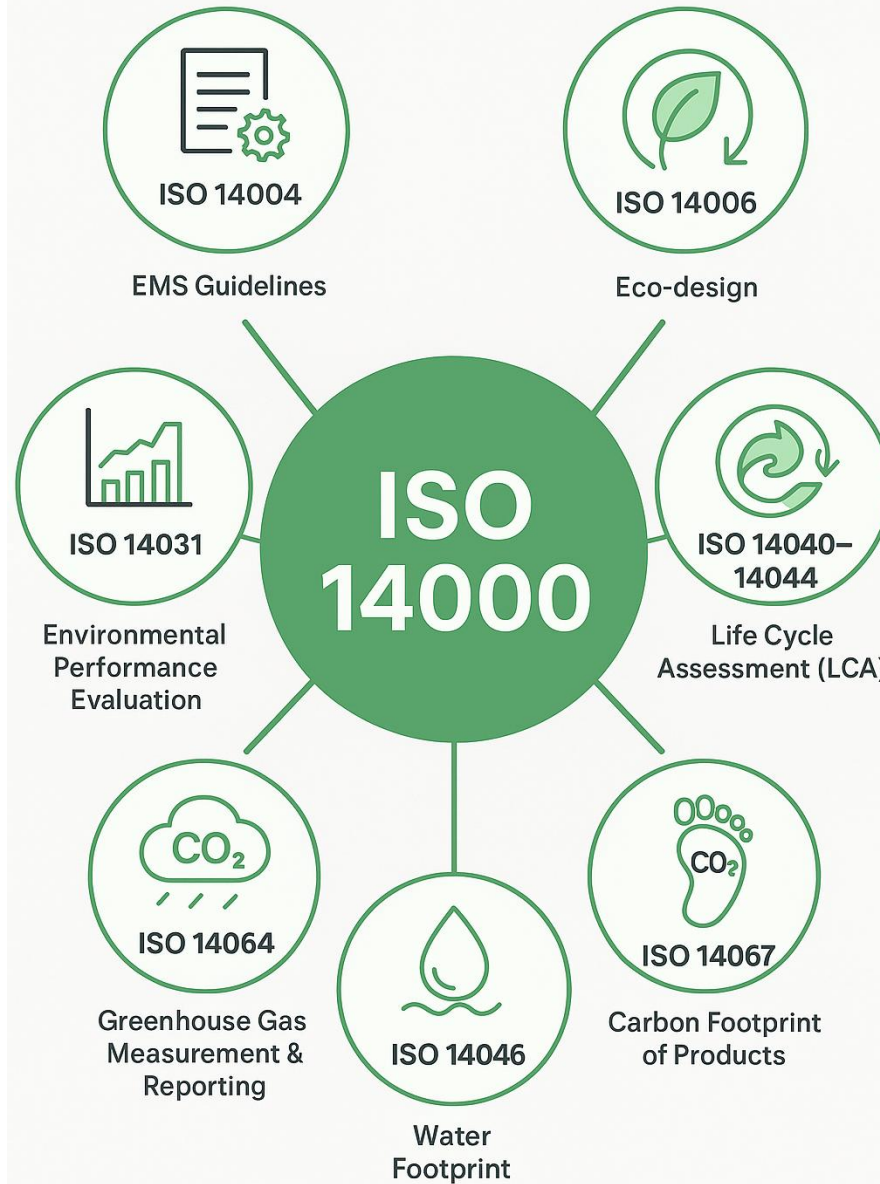
◆ Core Standard: ISO 14001

- ❖ **ISO 14001** is the main standard in the ISO 14000 family.
- ❖ It specifies requirements for an **Environmental Management System (EMS)**.
- ❖ It can be **applied to any organization, regardless of size, type, or sector**.

◆ Key Elements of ISO 14001 (EMS Framework)



Other Standards in ISO 14000 Family



ISO 14004 – EMS Guidelines

- ❑ Provides practical guidance for **establishing, implementing, maintaining, and improving** an Environmental Management System (EMS).

ISO 14006 – Eco-design

- ❑ Focuses on integrating **environmental considerations into product design & development**.
- ❑ Helps create sustainable products with **reduced environmental impact**.

ISO 14031 – Environmental Performance Evaluation

- ❑ Provides a framework to **monitor, measure, evaluate, and communicate** environmental performance.
- ❑ Uses performance indicators (emissions, resource use, waste).

ISO 14040–14044 – Life Cycle Assessment (LCA)

- ❑ Methodology to evaluate the **environmental impacts of products/services throughout their life cycle** (raw material → production → use → disposal).
 - ❑ ISO 14040: principles & framework.
 - ❑ ISO 14044: requirements & guidelines.

ISO 14064 – Greenhouse Gas (GHG) Measurement & Reporting

- ❑ Provides standards for **quantifying, monitoring, reporting, and verifying** greenhouse gas emissions and removals.

ISO 14067 – Carbon Footprint of Products

- ❑ Defines principles and methods for **quantifying and communicating the carbon footprint** of products.
- ❑ Useful for **carbon labeling** and climate action strategies.

ISO 14046 – Water Footprint

- ❑ Specifies principles and methods for **assessing water usage and impacts** associated with products, processes, and organizations.

Carbon labeling is the practice of displaying information about the **carbon footprint** (total greenhouse gas emissions) of a product, service, or activity on its label or packaging.

- ❑ It tells consumers how much **CO₂ equivalent (CO₂e)** was emitted during the **life cycle** of that product (production → transportation → use → disposal).

Linear Economy

1 Linear Economy

❑ **Model:** Take → Make → Use → Dispose

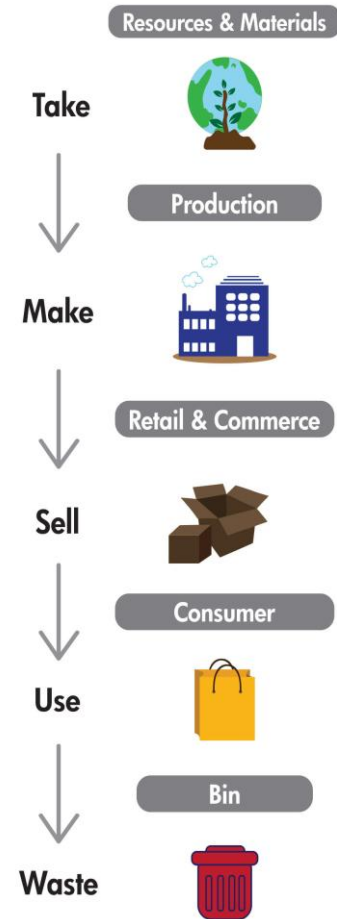
❑ **Process:**

- ❑ Extract raw materials
- ❑ Manufacture products
- ❑ Consumers use products
- ❑ Waste is thrown away (landfills, pollution)

❑ **Characteristics:**

- ❑ One-way flow of materials.
- ❑ Heavy reliance on virgin resources.

LINEAR ECONOMY



Linear economy - ISSUES

- The Industrial Revolution created a linear model for a rapid and an efficient production of products — more often with built-in service life — without regard to environmental consequences.
- Historically, manufacturers often took environmentally unsustainable resources, manufactured goods, and disposed of the remaining materials in ways that were harmful to the environment.
- The linear process. Usually, the process is optimized to make it as efficient as possible, with high throughput and lower production costs based on the availability of various ear economy is based on inputs at relatively low cost.

Circular Economy

2 Circular Economy

❑ **Model:** Reduce → Reuse → Recycle → Recover

❑ **Process:**

- ❑ Design products for durability, repair, and reuse.
- ❑ Extend product life through sharing, repairing, remanufacturing.
- ❑ Recycle materials back into production loops.
- ❑ Use renewable energy in production.

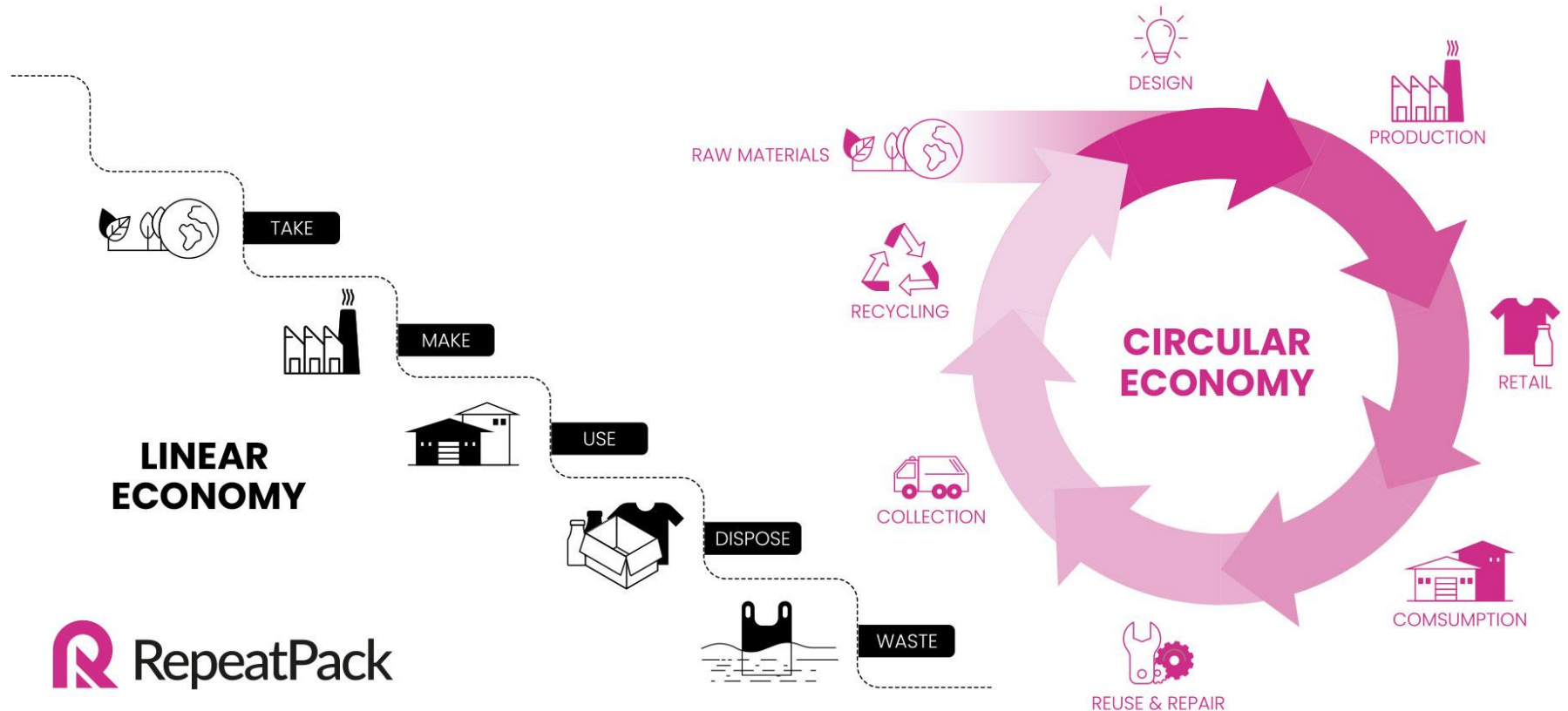
❑ **Characteristics:**

- ❑ Closed-loop system.
- ❑ Minimizes waste.
- ❑ Focuses on **resource efficiency and sustainability**.

❑ **Benefit:** Reduces environmental footprint, conserves resources, and promotes green innovation.



THE DIFFERENCE BETWEEN LINEAR AND **CIRCULAR** ECONOMIES



Aspect	Linear Economy 	Circular Economy 
Flow of resources	One-way (take–make–dispose)	Closed-loop (reuse–recycle–recover)
Resource use	Extract and consume	Optimize and regenerate
Waste	High (landfills, pollution)	Low (waste becomes resource)
Sustainability	Unsustainable	Sustainable
Business model	Sell more products	Provide services, reuse, eco-design
Example	Fast fashion industry	Clothing rental, recycling fabrics

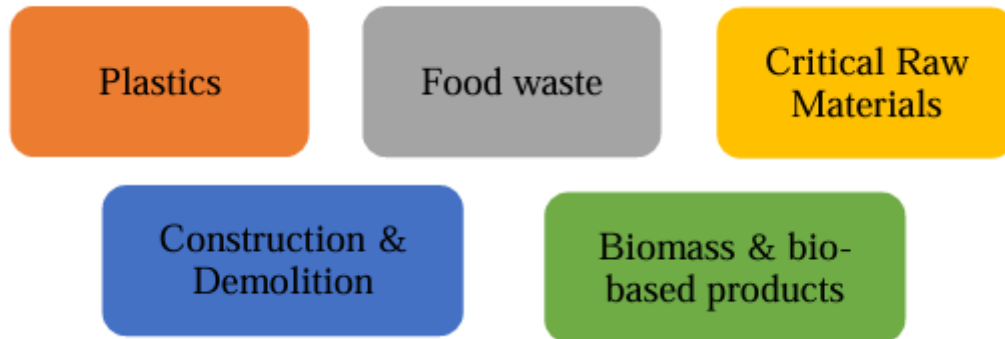
👉 In short:

❑ Linear economy = **throwaway culture** 

❑ Circular economy = **sustainability through closed loops**  

PRIORITY AREAS OF CIRCULAR ECONOMY

- Comprehensive **consultations** were held with various stakeholders, based on which the following **five priority sectors were identified**.
- All these sectors have their own problems and barriers on implementation and development.



Five priority areas of circular economy

Objectives and key actions of priority sectors of circular economy

Priority Area	Objectives	Key Actions
Plastics	❑ Increase recycling efficiency (currently <25% recycled, ~50% landfilled).	❑ Adopt a strategy to reduce plastic waste, including marine litter. ❑ Set higher recycling targets for plastic packaging in new legislation.
Food Waste	❑ Halve food waste by 2030 (SDG target). ❑ Currently ~100 million tons wasted annually in EU.	❑ Develop EU methodology to measure food waste. ❑ Create platform for sharing best practices. ❑ Improve understanding of date labeling (“best before”).
Critical Raw Materials (CRMs)	❑ Increase efficient use and recovery of scarce raw materials.	❑ Encourage recovery of CRMs. ❑ Draft report on best practices & options for EU action. ❑ Encourage Member States’ initiatives.

Objectives and key actions of priority sectors of circular economy

Priority Area	Objectives	Key Actions
Construction & Demolition	❑ Identify and recover valuable materials from building sector.	❑ Ensure recovery & waste management in construction/demolition. ❑ Assess environmental performance of buildings. ❑ Develop pre-demolition guidelines for high-value recycling. ❑ Promote voluntary recycling protocols.
Biomass & Bio-based Products	❑ Promote efficient use of wood & bio-based resources. ❑ Increase recycling of bio-waste.	❑ Encourage cascading use of biomass. ❑ Set new recycling targets for wood.

Key enabling factors on the way to the circular economy are



✓ In short: **Policy + Innovation + Markets + Infrastructure + Skills + Finance + Partnerships** are the main enabling pillars (**7 pillars**)

Key enabling factors on the way to the circular economy are

1. Policy and Regulation

- a. **Clear legislation and ambitious EU/national targets** for recycling, reuse, and waste reduction.
- b. **Incentives** for circular business models (e.g., **tax breaks, subsidies**).
- c. **Extended Producer Responsibility (EPR)** → making producers responsible for end-of-life management.

2. Innovation and Technology

- a. Advanced **recycling technologies** (mechanical, chemical, bio-based).
- b. **Digital tools**: product passports, blockchain for traceability, AI for waste sorting.
- c. **Eco-design & product innovation**: designing products for durability, reuse, and recyclability.

3. Market and Economic Incentives

- a. Creation of markets for **secondary raw materials**.
- b. New **business models**: product-as-a-service, sharing economy, repair and refurbishment industries.

4. Infrastructure and Logistics

- a. Efficient **collection, sorting, and recycling systems**.
- b. **Reverse logistics** for product returns and reuse.
- c. **Urban mining** (recovering resources from waste streams).

Key enabling factors on the way to the circular economy are

5. Knowledge, Skills, and Education

- a. Training workforce in circular skills (repair, remanufacturing, material science).
- b. Awareness campaigns to change consumer behavior (e.g., food waste, plastics).
- c. Academic–industry partnerships to drive R&D in circular economy solutions.

6. Finance and Investment

- a. Investment in green infrastructure and technologies.
- b. Access to sustainable finance (green bonds, EU funding programs).
- c. Risk-sharing mechanisms for companies adopting new circular models.

7. Collaboration and Partnerships

- a. Industry–government–research cooperation.
- b. Cross-sector collaborations to close resource loops (e.g., construction using industrial by-products).
- c. International cooperation for critical raw materials recovery.

**Product stewardship and
sustainable development goals
as guiding frameworks**

What is Product Stewardship?

❑ **Product Stewardship** is an environmental management approach where everyone involved in the life cycle of a product (**manufacturers, suppliers, retailers, consumers, and recyclers**) takes responsibility for minimizing the product's environmental impact .

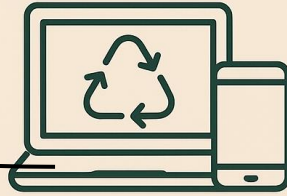
◆ Key Idea

👉 Responsibility is **shared**, but the **primary responsibility lies with the producer/manufacturer** to design products that are safe, sustainable, and recyclable → **Extended Producer Responsibility (EPR)**



Examples of product stewardship in action

Manufacturers take back old laptops/phones for recycling

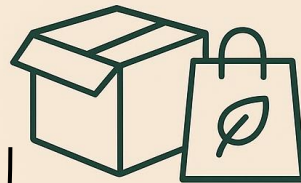


ELECTRONICS RECYCLING

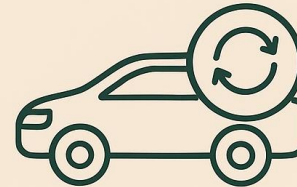
Companies provide collection systems for used batteries.



BATTERY DISPOSAL PROGRAMS



PACKAGING STEWARDSHIP



AUTOMOTIVE INDUSTRY

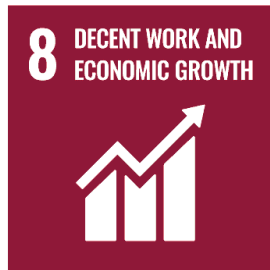
Car companies design vehicles with reusable parts/interchangeable and manage end-of-life scrapping

Brands use recyclable or biodegradable packaging and take responsibility for waste management

Sustainable Development Goals (SDG)

- ❑ **SDGs (Sustainable Development Goals)** are a set of 17 global goals adopted by all United Nations (UN) Member States in 2015 as part of the 2030 Agenda for Sustainable Development.
- ❑ They are a **blueprint for peace, prosperity, and sustainability**, addressing global challenges such as poverty, inequality, climate change, environmental degradation, and justice.

SUSTAINABLE DEVELOPMENT GOALS





SDG 1: NO POVERTY

End poverty in all its forms everywhere

WHY IT MATTERS

- Poverty affects access to food, healthcare, education, housing, and opportunities
- More than 700 million people worldwide live in extreme poverty (earning less than \$2.15 per day, World Bank, 2023).
- Poverty is closely linked with hunger, poor health, inequality, and conflict

EXAMPLE ACTIONS

- **Governments:** Provide financial aid, housing, and welfare schemes
- **Businesses:** Create decent jobs, pay fair wages, support local supplier
- **Communities:** Run skill-training centers, microfinance programs
- **Individuals:** Donate, volunteer, and support poverty alleviation initiatives

REAL-WORLD EXAMPLE

India's MGNREGA Scheme provides guaranteed rural employment for poor households

2 ZERO HUNGER



Goal Statement: End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.

Why it Matters:

- ❑ Over 735 million people face **chronic hunger** (2023).
- ❑ **Malnutrition** weakens health, education, and economic growth.
- ❑ **Sustainable farming** is essential for food security.

Key Targets:

- ❑ **End hunger and malnutrition for all, especially children.**
 - ❑ Double agricultural productivity of small-scale farmers.
 - ❑ Ensure sustainable food production and resilient farming systems.
 - ❑ Maintain genetic diversity of crops and livestock.

2 ZERO HUNGER



Example Actions:

- ☐ Promote school meal programs.
- ☐ Reduce food waste in supply chains.
- ☐ Support farmers with technology and fair trade.

Real-world Example:

- ☐ **School Meals Program** – provides nutritious meals to children in developing countries.

3 GOOD HEALTH AND WELL-BEING



Great strides have been made in improving people's health in recent years. As of 2023, 133 countries have already met the SDG target on under-5 mortality. Effective HIV treatment has cut global AIDS-related deaths by half since 2010 and at least one neglected tropical disease has been eliminated in 54 countries by the end of 2024.

However, inequalities in health care access persist. The COVID-19 pandemic and other ongoing crises have impeded progress towards Goal 3. Childhood vaccination rates have experienced a significant decline, and tuberculosis and malaria cases have increased, compared with pre-pandemic levels.

Goal Statement: Ensure healthy lives and promote well-being for all at all ages.

Why it Matters:

- ☐ Millions die each year from preventable diseases.
- ☐ Access to healthcare is unequal across the globe.
- ☐ Health is the foundation of productivity and prosperity.

Key Targets:

- ☐ Reduce maternal and child mortality.
- ☐ End epidemics (HIV, TB, malaria, neglected tropical diseases).
- ☐ Achieve universal health coverage and access to medicines/vaccines.
- ☐ Reduce deaths from road traffic accidents and substance abuse.

3 GOOD HEALTH AND WELL-BEING



Example Actions:

- ☐ Vaccination campaigns.
- ☐ Expanding health insurance coverage.
- ☐ Mental health support programs.

Real-world Example:

- ☐ **GAVI: the vaccine alliance** (is a global public-private partnership that aims to increase access to immunization in the world's poorest countries) – improving vaccine access in low-income countries.

4 QUALITY EDUCATION



Progress towards quality education was already slower than required before the COVID-19 pandemic further slowed gains. Global completion rates have risen only modestly since 2015 and learning proficiency remains low.

As of 2023, 272 million children and youth were out of school, up 3 per cent since 2015 and disproportionately concentrated in low-income countries.

In addition to free primary and secondary schooling for all boys and girls by 2030, the aim is to provide equal access to affordable vocational training, eliminate gender and wealth disparities, and achieve universal access to quality higher education.

Goal Statement: Ensure inclusive, equitable, and quality education and promote lifelong learning opportunities for all.

Why it Matters

- ☐ Education is the foundation for sustainable development and equality.
- ☐ Still, **244 million children and youth worldwide** are out of school (UNESCO, 2022).
- ☐ Without access to quality education, cycles of poverty, inequality, and poor health continue.
- ☐ Lifelong learning and skills are essential for decent jobs and innovation.

4 QUALITY EDUCATION



Example Actions

- ❑ Governments: Invest in schools, teacher training, and digital learning infrastructure.
- ❑ NGOs: Provide scholarships, literacy programs, and community schools.
- ❑ Businesses: Support skill-development, internships, and education funding.
- ❑ Individuals: Volunteer as tutors, donate books, promote digital literacy.

Real-world Example

- ❑ **Pratham (India)** – NGO working to improve literacy and learning outcomes for underprivileged children.
- ❑ **UNICEF's Education Programs** – supporting education access in conflict and crisis zones.

5 GENDER EQUALITY



Achieving gender equality and empowering all women and girls is not just the right thing to do, it's essential for sustainable development. Goal 5 focuses on closing the gender gap and ensuring equal opportunities for all.

Gender equality is not only a fundamental human right, but a necessary foundation for a peaceful, prosperous and sustainable world. There has been progress over the last decades, but the world is not on track to achieve gender equality by 2030.

Goal Statement: Achieve gender equality and empower all women and girls.

Why it Matters ?

- ☐ Women and girls represent **half of the world's population**, yet face systemic barriers to equality.
- ☐ Gender discrimination leads to limited access to education, healthcare, and job opportunities.
- ☐ Violence, child marriage, and unequal pay remain major global challenges.
- ☐ Empowering women boosts economic growth, innovation, and community development.

5 GENDER EQUALITY



Key Targets under SDG 5

- ☐ End all forms of **discrimination** against women and girls everywhere.
- ☐ **Eliminate violence, exploitation, trafficking, and harmful practices** (e.g., child marriage, female genital mutilation).
- ☐ Ensure **equal participation and leadership opportunities in political, economic, and public life**.
- ☐ Recognize and **value unpaid care and domestic work**.

Real-world Example

- ☐ **HeForShe Campaign (UN Women):**
A global movement engaging men and boys to support gender equality.
- ☐ **Self-Employed Women's Association (SEWA), India:**
Empowers women through microfinance, training, and collective bargaining.

6 CLEAN WATER AND SANITATION



Access to safe water, sanitation and hygiene is the most basic human need for health and well-being. Demand for water is rising owing to rapid population growth, urbanization and increasing water needs from agriculture, industry, and energy sectors.

The demand for water has outpaced population growth. Global water stress has held at 18 per cent since 2015, with one in ten people now living under high or critical stress and several regions exceeding 75 per cent stress. Water scarcity is projected to increase with the rise of global temperatures as a result of climate change.

Goal Statement: Ensure availability and sustainable management of water and sanitation for all.

Why it Matters

- ☐ **2.2 billion people** worldwide lack access to safe drinking water.
- ☐ **4.2 billion people** do not have safe sanitation services.
- ☐ Poor water and sanitation cause diseases like cholera and diarrhea, leading to millions of deaths annually, especially among children.

Key Targets under SDG 6

- ☐ Achieve universal and equitable access to safe and affordable drinking water.
- ☐ Ensure access to adequate sanitation and hygiene for all.
- ☐ Improve water quality by reducing pollution and minimizing hazardous chemicals.
- ☐ Increase water-use efficiency across all sectors (agriculture, industry, households).

6 CLEAN WATER AND SANITATION



Example Actions

- ☐ Governments: Build water infrastructure, regulate pollution, and fund sanitation programs.
- ☐ NGOs: Provide clean water systems, wells, and hygiene education in rural areas.
- ☐ Businesses: Reduce water use in manufacturing, treat wastewater, and invest in water-saving technologies.
- ☐ Individuals: Save water at home, avoid plastic pollution, and support clean water initiatives.

Real-world Example

- ☐ **Water.org (co-founded by Matt Damon):** Provides affordable financing for safe water and sanitation in developing countries.
- ☐ **Swachh Bharat Mission (India):** Nationwide program to eliminate open defecation and improve sanitation access.

7 AFFORDABLE AND CLEAN ENERGY



Goal 7 is about ensuring access to clean and affordable energy which is key to the development of agriculture, business, communications, education, healthcare and transportation.

The world continues to advance towards sustainable energy targets – but not fast enough. At the current pace, about 645 million people will still lack access to electricity and 1.8 billion people will still rely on polluting fuels and technologies for cooking by 2030.

Our everyday life depends on reliable and affordable energy. And yet the consumption of energy is the dominant contributor to climate change, accounting for two-thirds of total global greenhouse gas emissions.

From 2010 to 2023, the proportion of the global population with access to electricity has increased from 84 per cent to 92 per cent.

Goal Statement

Ensure access to affordable, reliable, sustainable, and modern energy for all.

◆ Why it Matters

- ❑ Around **675 million people still live without electricity** (IEA, 2023).
- ❑ **2.3 billion people** rely on wood, charcoal, or dung for cooking, causing health issues and deforestation.
- ❑ Energy access is critical for education, healthcare, industry, and quality of life.
- ❑ **Clean energy** reduces greenhouse gas emissions and combats climate change.

8 DECENT WORK AND ECONOMIC GROWTH



Promoting sustained economic growth, productive employment and decent work for all is essential for a sustainable future. Goal 8 focuses on creating quality jobs, promoting entrepreneurship, and fostering economic growth that benefits everyone.

The journey towards achieving Goal 8 faces significant challenges due to the aftermath of COVID-19, trade tensions, rising debt in developing countries and geopolitical conflicts. These factors collectively threaten global economic growth. Although global real GDP per capita saw a rebound in 2021, it experienced a slowdown in 2022 and is projected to slow to just 1.5 per cent in 2025 due to heightened trade tensions and policy uncertainty.

While the unemployment rate fell to a record low of 5.0 per cent in 2024, nearly 58 per cent of workers remained informally employed, with persistently high rates in LDCs and sub-Saharan Africa. However, significant disparities across gender and age persist.

◆ Goal Statement

Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.

◆ Why it Matters

- ☐ Around **2 billion people** work in the informal economy with limited rights and protection.
- ☐ Youth unemployment remains a major challenge worldwide.
- ☐ Economic growth that is not inclusive leads to inequality and social unrest.
- ☐ Sustainable jobs are essential for reducing poverty and improving living standards.

9 INDUSTRIES, INNOVATION AND INFRASTRUCTURE



Goal 9 seeks to build resilient infrastructure, promote sustainable industrialization and foster innovation. Goal 9 seeks to build resilient infrastructure, promote sustainable industrialization and foster innovation. Goal 9 seeks to build resilient infrastructure, promote sustainable industrialization and foster innovation.

Building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation are key drivers of economic development. Goal 9 aims to create a strong foundation for a sustainable future.

◆ Goal Statement

Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

◆ Why it Matters

- ❑ In developing regions, **poor infrastructure limits healthcare, education, and economic opportunities.**
- ❑ **Resilient industries and infrastructure** are critical to withstand crises such as pandemics and climate change.

10 REDUCED INEQUALITIES



Inequality threatens long-term social and economic development, harms poverty reduction and destroys people's sense of fulfillment and self-worth.

The incomes of the poorest 40 per cent of the population had been growing faster than the national average in most countries. But emerging yet inconclusive evidence suggests that COVID-19 may have put a dent in this positive trend of falling within-country inequality.

The global refugee population has surged to 37.8 million – primarily from Afghanistan, the Bolivarian Republic of Venezuela, the Syrian Arab Republic and Ukraine – intensifying humanitarian pressures.

◆ Goal Statement

Reduce inequality within and among countries.

◆ Why it Matters

- ☐ The richest **10% of the world's population** earns over half of global income.
- ☐ Inequalities fuel social tensions, migration pressures, and political instability.
- ☐ Discrimination based on gender, ethnicity, disability, or economic status restricts opportunities.
- ☐ Reducing inequality is essential for social justice, peace, and sustainable growth.

11 SUSTAINABLE CITIES AND COMMUNITIES



Cities are centers of innovation and economic growth, but they also face challenges like poverty, inequality and environmental degradation. Goal 11 promotes sustainable urban development to create inclusive, safe, and resilient cities.

Cities represent the future of global living. The world's population reached 8 billion in 2022 with over half living in urban areas. This figure is only expected to rise, with 70 per cent of people expected to live in cities by 2050.

However many of these cities are not ready for this rapid urbanisation, and it outpaces the development of housing, infrastructure and services, which led to a rise in slums or slum-like conditions.

◆ Goal Statement

Make cities and human settlements inclusive, safe, resilient, and sustainable.

◆ Why it Matters

- ❑ Over **55% of the world's population** lives in cities, projected to reach **68% by 2050**.
- ❑ Urban areas consume **over 70% of global energy** and produce most greenhouse gas emissions.
- ❑ Rapid urbanization leads to overcrowding, pollution, poor housing, and infrastructure strain.
- ❑ Safe and sustainable cities are crucial for human well-being and climate resilience.

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



Goal 12 is about ensuring sustainable consumption and production patterns, which is key to sustain the livelihoods of current and future generations.

Our planet is running out of resources, but populations are continuing to grow. If the global population reaches 9.8 billion by 2050, the equivalent of almost three planets will be required to provide the natural resources needed to sustain current lifestyles.

As of 2024, 530 policies related to sustainable consumption and production were submitted across 71 countries, a 6 per cent increase from 2023. We need to change our consumption habits, and shifting our energy supplies to more sustainable ones is one of the main changes we must make if we are going to reduce our consumption levels.

◆ Goal Statement

Ensure sustainable consumption and production patterns.

◆ Why it Matters

- ❑ The world's population is expected to reach **9.7 billion by 2050**, putting pressure on resources.
- ❑ **Over 1/3 of all food produced** is wasted, while millions still go hungry.
- ❑ Unsustainable consumption drives pollution, biodiversity loss, and climate change.
- ❑ Shifting to sustainable production reduces costs, saves resources, and protects the planet.

13 CLIMATE ACTION



Every person, in every country in every continent is impacted in some shape or form by climate change. Climate change is caused by human activities – primarily the burning of fossil fuels like oil, coal and gas – and threatens life on Earth as we know it. With rising greenhouse gas emissions, the impacts of climate change are intensifying and accelerating, including more frequent extreme weather events like droughts, floods and storms, and rising sea levels.

If left unchecked, climate change will undo a lot of the development progress made over the past years. It can also intensify conflicts over resources and force people to move.

Currently, our planet is already about 1.2°C warmer than it was in the late 1800s, and emissions continue to rise. To keep global warming to no more than 1.5°C – as called for in the Paris Agreement – greenhouse gas emissions must be cut by almost half by 2030 compared to 2019.

◆ Goal Statement

Take urgent action to combat climate change and its impacts.

◆ Why it Matters

- ☐ The past eight years have been the **warmest on record** globally.
- ☐ Climate change causes extreme weather events, rising sea levels, and biodiversity loss.
- ☐ Vulnerable populations in developing countries are disproportionately affected.
- ☐ Immediate action is essential to limit global warming to **1.5°C above pre-industrial levels**.

14 LIFE BELOW WATER



Goal 14 is about conserving and sustainably using the oceans, seas and marine resources. A healthy ocean and seas are essential to human existence and life on Earth.

The Ocean is intrinsic to our life on earth. Covering three-quarters of the Earth's surface, contain 97 percent of the Earth's water, and represent 99 percent of the living space on the planet by volume.

It provides key natural resources including food, medicines, biofuels and other products; help with the breakdown and removal of waste and pollution; and their coastal ecosystems act as buffers to reduce damage from storms. It also acts as the planet's greatest carbon sink.

◆ Goal Statement

Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.

◆ Why it Matters

- ☐ Oceans cover **70% of the Earth's surface** and provide food, jobs, and oxygen.
- ☐ Over **30% of fish stocks** are overexploited, threatening global food security.
- ☐ Marine pollution (especially plastics) kills over **100,000 marine animals** each year.
- ☐ Healthy oceans regulate climate, absorb carbon dioxide, and support biodiversity.

15 LIFE ON LAND



Goal 15 is about conserving life on land. It is to protect and restore terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and stop biodiversity loss.

Earth's ecosystems are vital for sustaining human life, they contribute to over half of global GDP and encompass diverse cultural, spiritual, and economic values.

However, the world is facing a triple crisis of climate change, pollution and biodiversity loss.

◆ Goal Statement

Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation, and stop biodiversity loss.

◆ Why it Matters

- ❑ Around **1 million species** face the risk of extinction due to human activities.
- ❑ Forests cover **31% of Earth's land area**, but deforestation continues at alarming rates.
- ❑ Land degradation affects **1.9 billion hectares** globally, reducing food security.
- ❑ Healthy ecosystems provide clean air, water, fertile soil, and are vital for climate regulation.

16 PEACE, JUSTICE AND STRONG INSTITUTIONS



Goal 16 is about promoting peaceful and inclusive societies, providing access to justice for all and building effective, accountable and inclusive institutions at all levels. People everywhere should be free of fear from all forms of violence and feel safe as they go about their lives whatever their ethnicity, faith or sexual orientation.

However, ongoing and new violent conflicts around the world are derailing the global path to peace and achievement of Goal 16. Alarmingly, the year 2022 witnessed a more than 50 per cent increase in conflict-related civilian deaths – the first since the adoption of Agenda 2030 – largely due to the war in Ukraine. In 2024, loss of lives amid armed conflicts surged 40 per cent from 2023, marking the third consecutive annual rise. About four times more children and women were killed in 2023–2024 than in the previous biennium; of these, 8 in 10 child deaths and 7 in 10 female deaths occurred in Gaza.

◆ Goal Statement

Promote peaceful and inclusive societies, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels.

◆ Why it Matters

- ☐ In 2022, **over 108 million people** were forcibly displaced by conflict, violence, and persecution.
- ☐ Corruption, weak governance, and injustice undermine development efforts.
- ☐ Peace, justice, and strong institutions are the foundation for achieving all other SDGs.
- ☐ Societies with inclusive governance and rule of law experience more stability and prosperity.

17 PARTNERSHIPS FOR THE GOALS



Goal 17 is about revitalizing the global partnership for sustainable development. The 2030 Agenda is universal and calls for action by all countries – developed and developing – to ensure no one is left behind. It requires partnerships between governments, the private sector, and civil society.

The Sustainable Development Goals can only be realized with a strong commitment to global partnership and cooperation to ensure no one is left behind in our journey to development.

However, not all countries are setting off from the same start line, and low and middle income countries are facing a tidal wave of debt which they are treading water.

◆ **Goal Statement**

Strengthen the means of implementation and revitalize the global partnership for sustainable development.

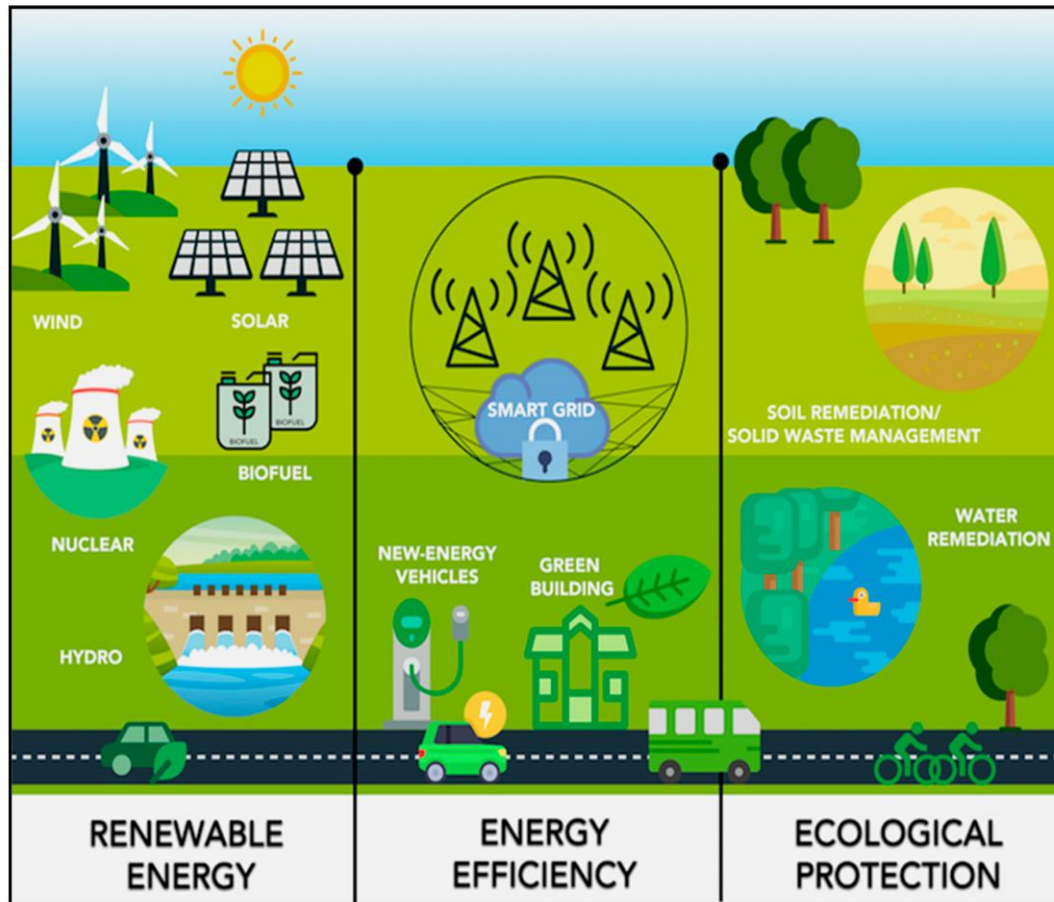
◆ **Why it Matters**

- ☐ No single country can achieve the SDGs alone — global cooperation is essential.
- ☐ Developing nations often lack resources, technology, and financing for sustainable growth.
- ☐ Strong partnerships between governments, businesses, NGOs, and citizens drive innovation and progress.
- ☐ Shared knowledge, trade, and technology can accelerate achievement of all SDGs.

SDG: <https://www.youtube.com/watch?v=0XTBYMfZyrM>

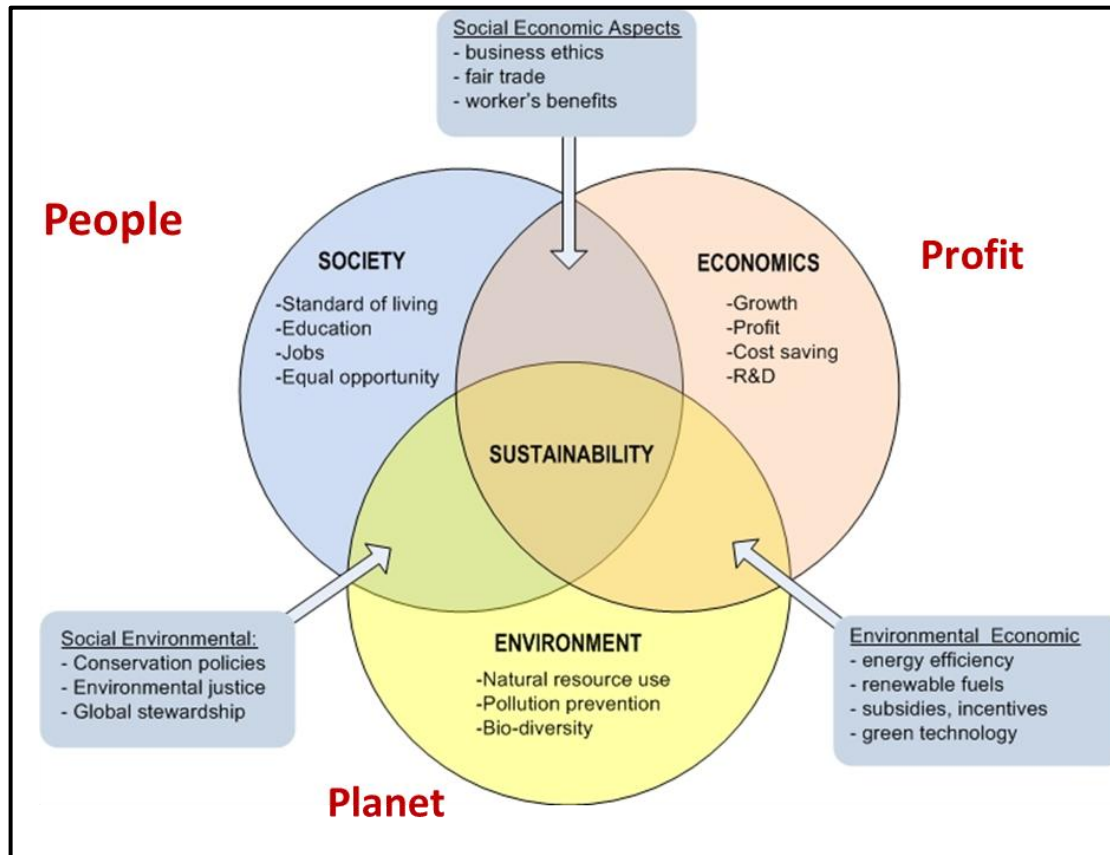
Role of Engineers in Sustainable Transitions (1/8)

1. Innovators of Green Technology – Develop renewable energy systems, low-emission manufacturing processes, and sustainable materials.



Role of Engineers in Sustainable Transitions (2/8)

2. Integrators of Systems Thinking – Consider environmental, social, and economic impacts together when designing projects.



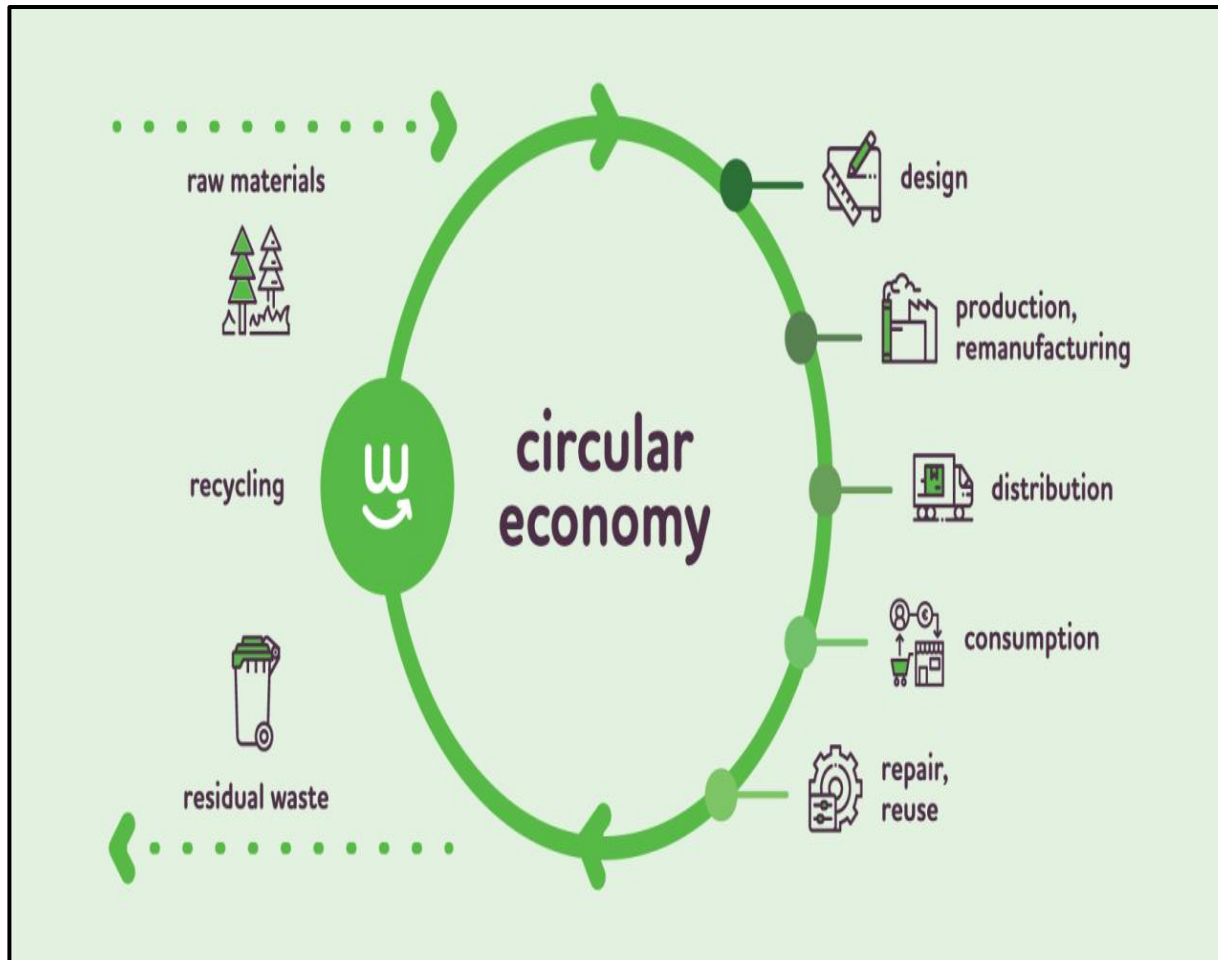
Role of Engineers in Sustainable Transitions (3/8)

3. Lifecycle Planners – Apply **lifecycle assessment (LCA)** to minimize total environmental footprint from raw material extraction to end-of-life.



Role of Engineers in Sustainable Transitions (4/8)

4. Enablers of Circular Economy – Design products for reuse, repair, remanufacture, and recycling



Role of Engineers in Sustainable Transitions (5/8)

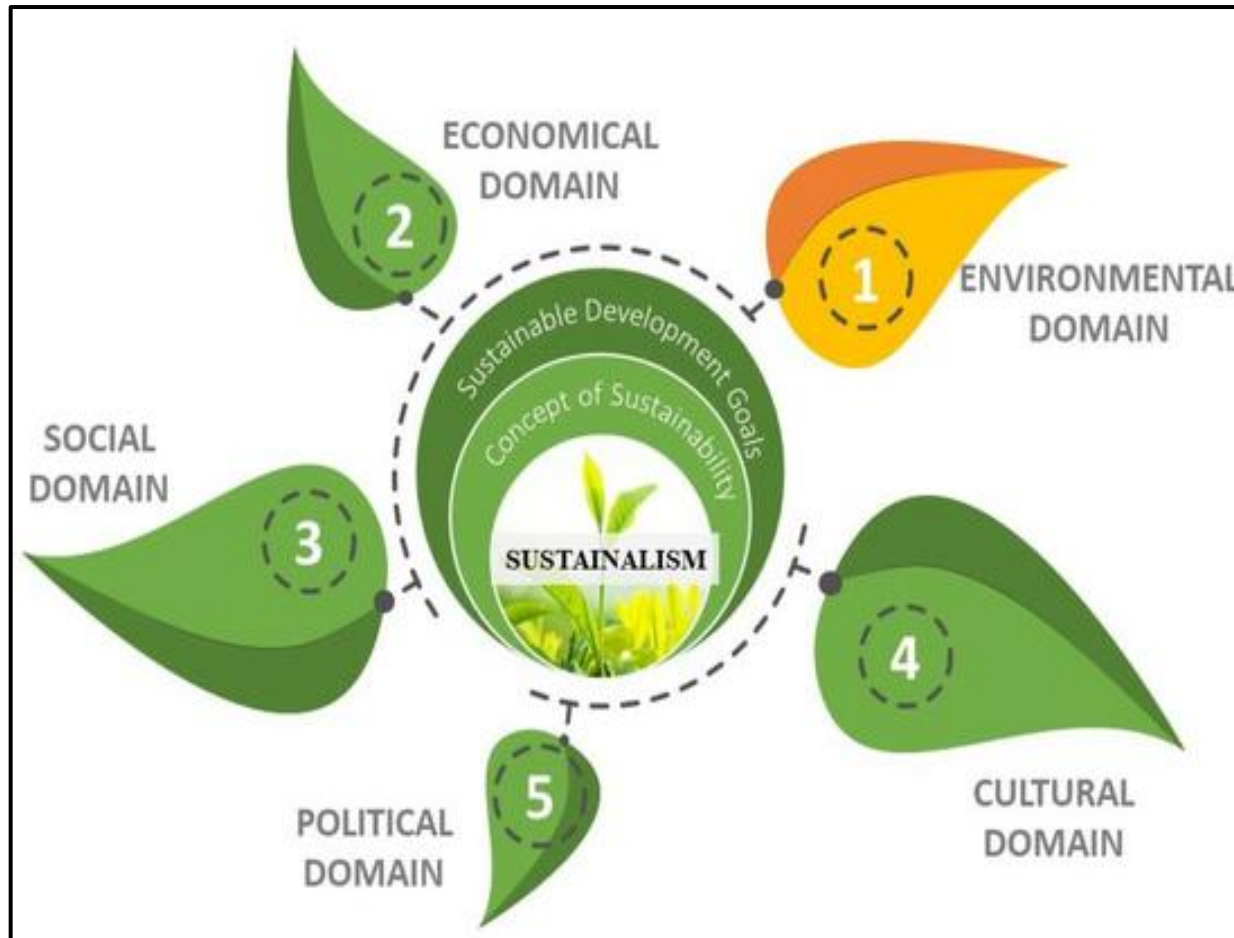
5. Resource Efficiency Champions – Optimize processes to use less energy, water, and raw materials without compromising quality.

Energy and Resource Efficiency Measures

-
- 01 Energy Audits and Monitoring
 - 02 Lighting Upgrades and Automation
 - 03 Equipment and Appliance Efficiency
 - 04 Water Conservation
 - 05 Supply Chain Optimization
 - 06 Workspace Design and Layout
 - 07 Employee Awareness and Training
 - 08 Renewable Energy Sources
 - 09 Waste Reduction and Recycling
 - 10 Collaboration and Benchmarking

Role of Engineers in Sustainable Transitions (6/8)

6. Policy & Standard Influencers – Collaborate with governments, industry bodies, and communities **to set and follow sustainability standards.**



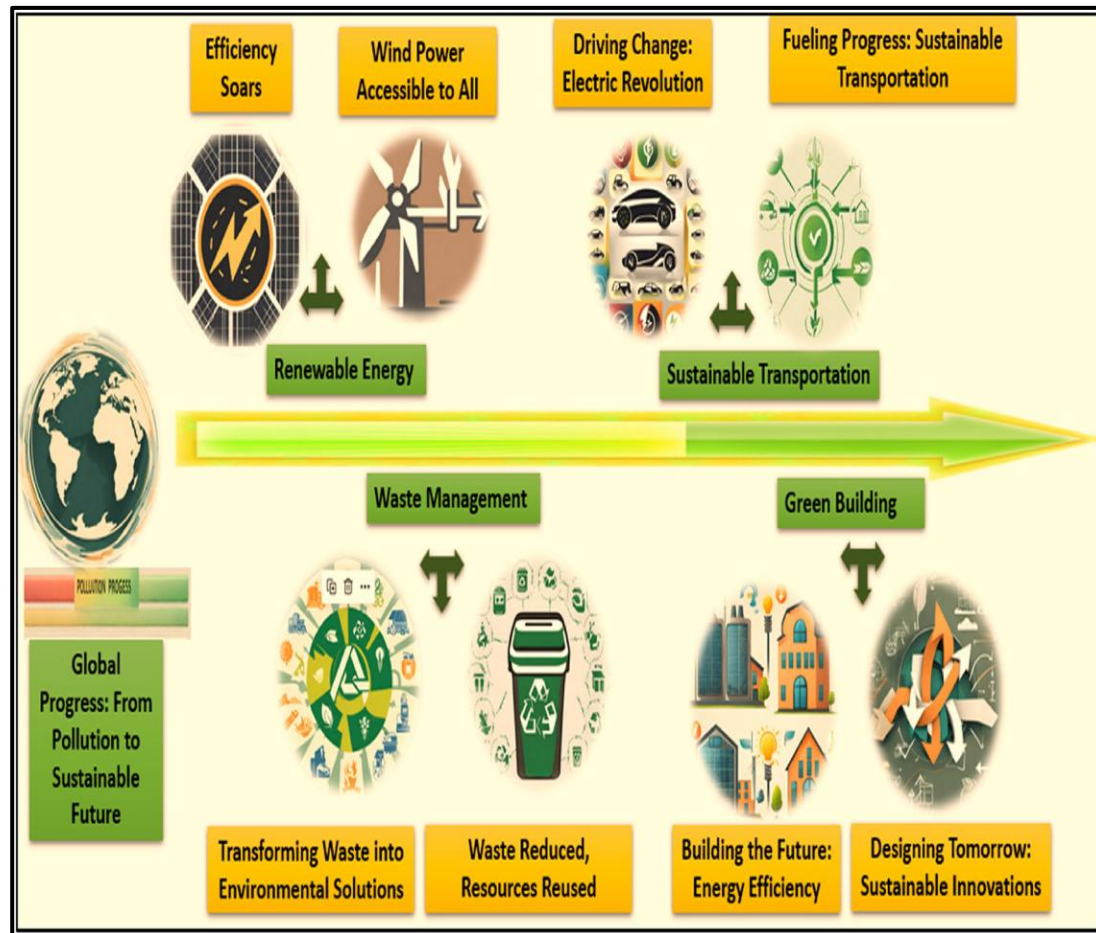
Role of Engineers in Sustainable Transitions (7/8)

7. Educators & Change Agents – Promote sustainable practices within teams, organizations, and society



Role of Engineers in Sustainable Transitions (8/8)

8. Sector-Specific Problem Solvers – Adapt sustainable solutions to unique needs of sectors such as **energy**, **transport**, **construction**, **agriculture**, and **healthcare**.



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